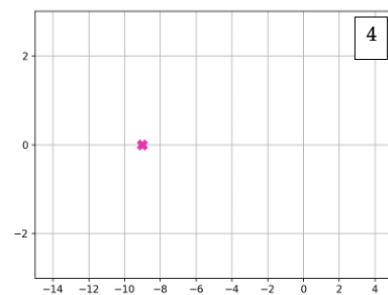
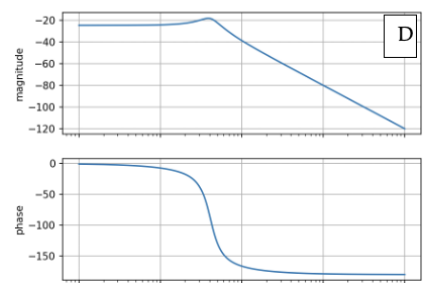
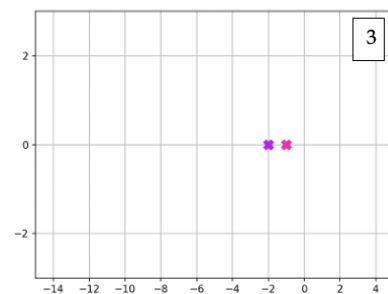
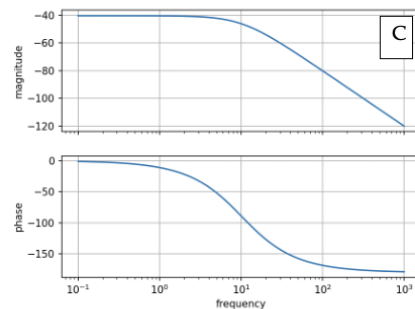
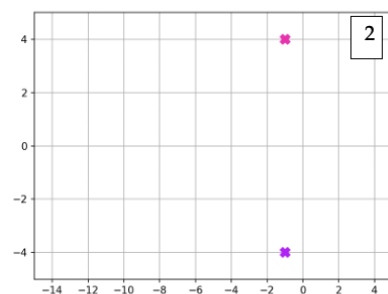
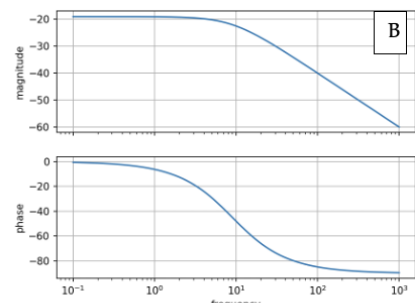
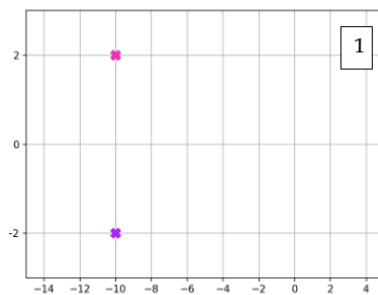
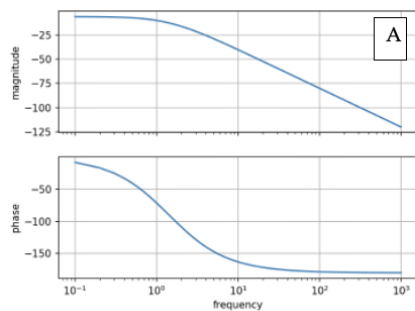


University of Cape Town
Department of Electrical Engineering
EEE3094S - Control Systems Engineering

Problem Set 3.1 Bode Plots

Question 1

Match the following Bode plots to the poles and zeros of the system:



Question 2

Sketch rough Bode plots for the following systems:

(a) $\frac{1}{s + 0.1}$

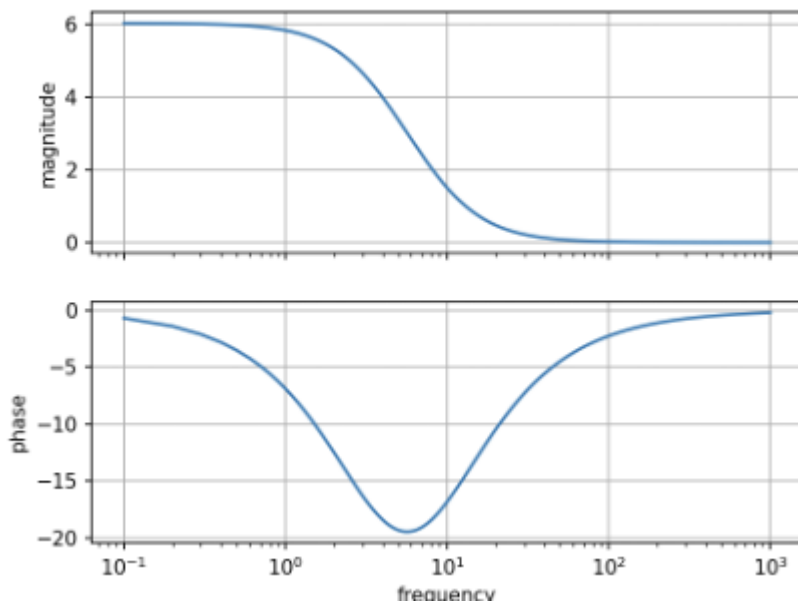
(b) $\frac{s + 5}{s + 50}$

(c) $\frac{1}{s^2 + 16s + 68}$

(d) $\frac{1}{(s + 10)(s + 12)(s + 20)}$

Question 3

The following Bode plot shows the frequency response of a first-order system.



- (a) Does this system have a zero, and if so, is it to the right or left of the pole? Explain.
- (b) What is the gain of the transfer function?
- (c) Would you recommend using this system in an environment where it will be exposed to high-frequency noise? Explain.